

VAL' KARKAY, Russian Federation

WMO: 219780

Lat: 70.084N

Lon: 170.925E

Elev: 19

StdP: 14.69

Time Zone: 12.00 (E12)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.3	-34.3	-40.9	0.5	-36.9	-38.2	0.6	-34.1	49.9	7.2	43.4	13.7	4.9	210	1.095

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	64.6	54.2	58.7	51.0	52.9	47.6	55.0	63.5	51.4	58.4	47.8	52.7	22.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
49.8	53.0	58.5	46.6	47.0	54.7	43.6	41.9	50.2	23.2	64.1	21.1	58.5	19.1	52.8	62.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
35.5	30.2	26.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			-37.7	57.2	3.6	4.8	-40.3	60.7	-42.4	63.5	-44.4	66.2	-47.0	69.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	-12.5	-11.0	-10.0	7.7	23.1	36.1	40.9	39.6	35.0	22.1	7.3	-7.7
	DBStd	22.92	12.79	14.04	12.88	10.96	8.08	6.50	7.17	6.57	5.97	10.28	13.55	14.93
	HDD50	13058	1937	1707	1859	1268	835	423	305	333	455	865	1281	1790
	HDD65	18487	2402	2127	2324	1718	1300	866	747	786	901	1330	1731	2255
	CDD50	46	0	0	0	0	0	7	23	12	4	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.7	12.6	11.8	9.6	10.0	10.6	11.7	12.4	12.8	12.2	10.9	12.7	12.5
Precipitation	PrecAvg	8.4	0.4	0.4	0.4	0.3	0.4	0.7	1.1	1.4	1.3	1.1	0.7	0.5
	PrecMax	11.2	0.8	0.9	1.0	0.8	1.0	1.5	1.9	2.4	2.7	3.3	1.6	1.0
	PrecMin	6.2	0.1	0.0	0.0	0.1	0.0	0.2	0.4	0.7	0.4	0.3	0.3	0.1
	PrecStd	1.4	0.2	0.2	0.2	0.2	0.2	0.3	0.4	0.5	0.5	0.6	0.3	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	33.3	35.3	37.2	42.2	68.9	72.3	65.9	61.6	44.2	36.1	34.7
		MCWB	30.5	31.4	32.5	33.1	35.2	54.9	58.7	54.4	53.5	40.7	32.8	31.9
	2%	DB	28.3	23.7	25.0	33.4	38.4	61.2	66.3	59.7	52.9	38.6	31.9	24.8
		MCWB	25.3	22.0	23.3	30.1	34.8	50.9	55.7	52.5	47.5	36.8	30.1	23.6
	5%	DB	10.8	15.0	14.2	29.4	35.7	53.2	60.3	55.1	46.3	36.6	28.8	19.5
		MCWB	9.9	14.1	13.1	27.0	33.2	46.2	52.6	49.2	42.3	35.6	27.1	18.4
	10%	DB	2.0	7.2	8.5	24.8	33.4	45.4	52.7	50.3	42.0	34.1	25.2	14.0
		MCWB	1.7	6.8	7.8	23.5	32.1	40.9	48.1	47.1	40.0	33.3	23.8	13.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.9	31.4	32.2	32.9	36.9	55.1	59.4	55.8	53.5	41.0	32.9	32.1
		MCDB	33.2	33.2	35.3	36.7	39.8	68.1	69.9	61.9	59.6	43.2	35.2	34.6
	2%	WB	25.8	21.8	23.4	30.3	34.8	50.9	56.7	52.7	47.6	37.4	30.5	23.3
		MCDB	28.1	23.7	25.4	33.2	37.4	60.9	65.2	59.1	52.9	38.0	32.0	24.9
	5%	WB	10.3	14.3	13.0	27.2	33.5	45.9	52.6	49.7	43.0	35.4	27.5	18.4
		MCDB	11.0	15.1	14.0	29.4	35.7	52.9	60.6	54.2	45.6	36.4	28.7	19.4
	10%	WB	1.8	6.8	7.8	23.4	32.2	40.9	47.9	46.7	40.3	33.3	23.7	13.3
		MCDB	2.1	7.1	8.5	25.0	33.6	45.0	52.5	50.1	42.0	34.1	25.2	14.1
Mean Daily Temperature Range	5% DB	MDBR	9.8	9.6	11.8	12.8	8.5	8.3	9.2	7.6	6.0	7.6	9.8	9.1
		MCDBR	20.5	16.9	17.4	13.9	10.7	22.6	22.8	18.6	13.7	7.3	10.5	12.7
		MCWBR	19.1	15.6	16.2	12.1	8.4	14.0	14.3	12.2	8.9	6.1	9.3	12.0
	5% WB	MCDBR	20.7	16.0	17.4	13.3	8.6	21.9	21.8	16.9	12.7	6.3	10.5	12.6
		MCWBR	19.2	15.4	16.2	11.7	7.0	13.9	14.0	11.5	8.7	5.4	9.8	11.9
Clear-Sky Solar Irradiance	taub	N/A	0.207	0.283	0.324	0.349	0.346	0.370	0.373	0.311	0.245	N/A	N/A	
	taud	N/A	2.016	2.027	1.974	2.025	2.259	2.334	2.337	2.460	2.124	N/A	N/A	
	Ebn at Noon	0	167	221	249	259	266	255	237	225	143	0	0	
	Edh at Noon	0	12	27	39	43	35	32	28	18	11	0	0	
All-Sky Solar Radiation	RadAvg	0	103	622	1382	1927	2064	1637	1039	516	147	0	0	
	RadStd	0	16	35	63	109	166	131	94	37	17	0	0	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page